Discrete Resonance Tuning as Optimal Quantization

Design Laws for the Number of Tuning States in Electromagnetic and Piezoelectric Vibration Energy Harvesters

Amey Mishra · Harmony School of Advancement, Houston, Texas · **Computational study: closed-form laws + exact dynamic-programming validation**

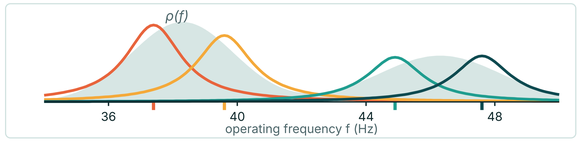
1 The problem

Machines change speed; a resonant harvester does not

- Condition-monitoring nodes on pumps, fans and compressors can run on the machine's own vibration instead of batteries [1]
- A linear harvester gives useful power only within a loaded bandwidth of a few per cent of its resonance [1]; a drive that changes speed leaves it **off resonance**
- **Discrete tuning states** (swapped or latched proof masses) avoid continuous actuation, but no theory says **how many to build or where**

2 The idea

Choosing N states for a speed distribution is optimal quantization



Schematic: each state is a Lorentzian of width $\Delta f_{1/2}$; N states cover the power-weighted speed density ρ(f).

- **States** = reproduction points · **ρ(f)** = source density · **1 − L** = distortion
- EM and PZ harvesters obey the **same laws** through a damping ratio γ and an area factor Γ

$$\varphi_N = \int \rho(f)\, \max_s L\left(\frac{2(f-f_s)}{\Delta f_{1/2}(f)}\right) df$$

$$\rho = \frac{qP_{\max}}{\int qP_{\max}\, df}, \qquad L(u) = \frac{1}{1+u^2}$$

Reference design (declared values, not measurements)

Steel cantilever	40 × 15 × 0.5 mm, k = 1464.8 N/m
20 g tip mass; 0.1 g RMS base	f_n = 42.49 Hz
EM transducer	Q = 25, R_c = 57 Ω, L_c = 30 mH, θ = 2.5 V·s/m

3 The question

How many states N, where, and what controls the answer?

- Depends on the speed distribution, Q, coupling and transducer type
- Prior work covers broadband random vibration [3,4], simultaneous arrays [7] and nonlinear broadening [9], not **switched states under multi-speed excitation**
- **Study type:** computational; no prototype was built

4 How it was tested

- **Full single-DOF models** (coil inductance, piezo capacitance) give every state's power in every speed bin
- **Dynamic programming** over contiguous cells gives the exact optimum for N = 1–30
- **8,400 random profiles:** 2,400 continuous (six shapes, Q = 10–60) and 6,000 discrete (K = 2–8)
- **Recorded data:** 49 MaFaulDa and 4 CWRU recordings, 40 strongest lines each [8]

Numerical checks on the optimizer

- Continuous refinement of each state changed power by ≤ 0.75%
- Unrestricted differential evolution beat the contiguous design by ≤ 0.008% (EM), ≤ 0.032% (PZ)

Code and data (Zenodo)
doi:10.5281/zenodo.23291174
Python, NumPy, SciPy; fixed seeds

BOTTOM LINE

Count states in bandwidths, not hertz: sparse states are capped by a Q-independent power quantum; dense states lose Λ²/(3N²)

5 · RESULTS: FROM PHYSICS TO A DESIGN RULE

1 Unify — One description for EM and PZ? **EM within 1.2%, PZ within 7.3%**

2 Few states — What caps sparse designs? **A per-state quantum, no Q**

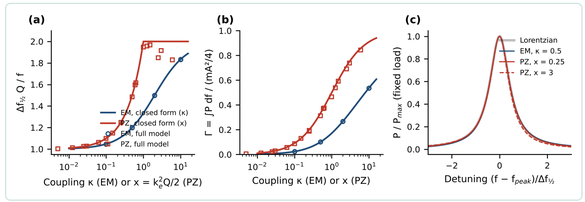
3 Many states — How fast does loss fall? **1/N² with cube-root placement**

4 Predict N — Can N be found quickly? **88–99% exact**

STEP 1

Unify EM and PZ

Yes: two numbers, γ and Γ, give bandwidth, power and best load for both.



(a) Bandwidth and (b) area factor vs coupling: closed forms (lines), full models (markers). (c) Loaded response is Lorentzian.

- EM forms exact without inductance [2]; with it, within 1.13%. PZ forms leading-order [6]: peak power within 1.25%, bandwidth 2.1%, integral 7.3%
- Frequency-integrated power is **independent of Q** and of f at fixed mass
- PZ mass tuning: open-circuit-side load branch gives **+9%, +29%, +46%** power at x = 1.25, 2.5, 3.75

TECHNICAL DETAIL

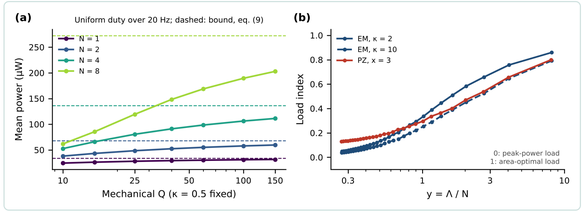
$$\gamma = \frac{c_e^*}{c} = \begin{cases} \kappa/(2+\kappa) & \text{EM} \\ \min(x,1) & \text{PZ} \end{cases} \qquad \Delta f_{1/2} = \frac{f}{Q}(1+\gamma)$$

$$\Gamma = \frac{\sqrt{1+\kappa}-1}{\sqrt{1+\kappa}+1}\ \text{(EM)}, \qquad \Gamma = \frac{x}{1+x}\ \text{(PZ)}$$

STEP 2

Few states

Each state adds at most ΓmA² max q / 4, whatever Q.



(a) Mean power vs Q, N-state optima (dashed: bound). (b) Optimal load moves from peak-power (0) to area-optimal (1) as Λ/N grows.

- **N = 1:** Q up 15× lifts power only 24.4 → 31.5 μW (92.5% of the 34.1 μW bound); **N = 8:** 61.7 → 202.9 μW
- **Equal-mass arrays** of 2–6 resonators give 34.6–35.9 μW, below the best single state (39.9); **switching** 2 and 3 states gives 68.6 and 90.5 μW

TECHNICAL DETAIL

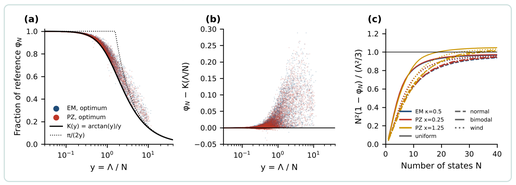
$$\bar{P}_N \le \sum_{s=1}^{N} \max_f q \int P_s\, df \le N\,\frac{\Gamma\, mA^2}{4}\,\max_f q(f)$$

- Scaling m, c, k, θ² by 1/N scales power by exactly 1/N: an array cannot beat one state
- Area-optimal EM load $R_c\sqrt{(1+\kappa)}$ = 69.8 Ω vs 71.2 Ω numerically

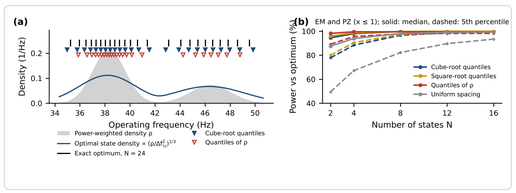
STEP 3

Many states

Loss falls as Λ²/(3N²); place states at cube-root quantiles.



(a) 2,400 cases collapse onto K(Λ/N); (b) its error; (c) convergence to Λ²/(3N²).



(a) Exact optimum (N = 24) follows the cube-root density; (b) closed-form placements.

- $N^2(1-\varphi_N)/(\Lambda^2/3)$ = 0.94–1.05 at N = 40
- Cube-root quantiles, no optimization: median **94.5, 98.0, 99.7, 99.98%** of the optimum at N = 2, 4, 8, 16 [5]

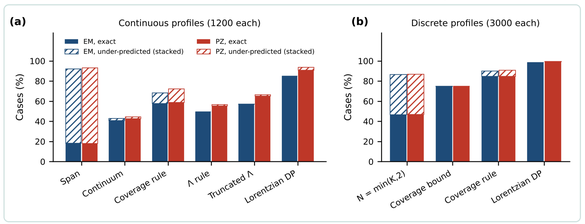
TECHNICAL DETAIL

$$1-\varphi_N \approx \frac{\Lambda^2}{3N^2}, \qquad \Lambda = \left(\int \left(\frac{\rho}{\Delta f_{1/2}^2}\right)^{1/3} df\right)^{3/2}, \qquad \lambda^* \propto \left(\frac{\rho}{\Delta f_{1/2}^2}\right)^{1/3}$$

STEP 4

Predict N

A closed form that rarely under-predicts, or a tiny nominal DP.



Predictors of N_{90}: exact (solid) or under-predicting (hatched).

Recorded scenario, N_{90}	EM	PZ x=.25	PZ x=1.5
R1 · 6 set points, 36.7–46.5 Hz	5	5	4
R2 · 30 speeds, 31–60 Hz	18	17	13
R3 · 2 speeds (60/40)	2	2	2
R4 · CWRU motor, 155–162 Hz	2	2	1

- **Nominal Lorentzian DP:** exact in 88% (EM), 92% (PZ) of continuous and 98.6–99.3% of discrete cases, vs 47% for the best constant guess; exact in 35 of 36 recorded comparisons
- **Truncated Λ rule:** exact in 60–66%, under-predicts 0.2–1.3%

TECHNICAL DETAIL

$$\varphi_N \simeq K\left(\frac{\Lambda}{N}\right), \quad K(y) = \frac{\arctan y}{y}, \quad N_\alpha = \left\lceil \frac{\Lambda}{y_\alpha} \right\rceil, \quad K(y_\alpha) = \alpha$$

$$y_{0.8} = 0.949, \quad y_{0.9} = 0.603, \quad y_{0.95} = 0.406$$

6 So, does it work?

In simulation: N is set by bandwidths, not by Q or span alone

- **Sparse:** N ≥ 4P̄*/(ΓmA² max q); no choice of Q beats it
- **Dense:** loss falls as 1/N², states at cube-root quantiles of $\rho/\Delta f^2_{1/2}$
- **Switching beats staggering:** an equal-mass array cannot exceed one state's power
- **Benchmark (five set points, 40–45 Hz):** two states raised mean power **49%** (62.07 → 92.70 μW)
- **Wide sweeps** (30 speeds) need 13–18 states for 90%: continuous tuning is better

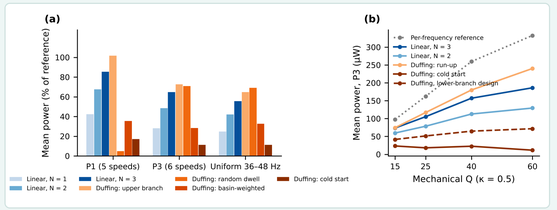
7 How sure are these results?

Check	Cases	Result
Unrestricted search beats DP design by	120	≤ 0.032%
Compander K(y) overestimate > 0.005	72,000	0.05–0.2%
Truncated Λ rule under-predicts	2,400	0.2–1.3%
Nominal DP exact (discrete)	6,000	98.6–99.3%
DP exact on recorded spectra	36	35 exact

Tolerances and per-unit calibration (EM)

- Stiffness, coil resistance, coupling ±5%, Q ±20%, mass ±1%; 100,000 simulated devices
- Uncalibrated: optimizing device-level 10th-percentile power gains **3.6–5.1%**; after one frequency measurement the deterministic design is within **0.12%**
- Predictor accuracy falls 10–20 points but the ranking holds

Versus a hardening (Duffing) harvester (EM, κ = 0.5)



(a) Mean power, % of per-frequency reference. (b) Power vs Q on profile P3.

- **Upper branch only:** 157.7 vs 132.5 μW for three linear states (P1)
- **Basin-weighted:** 55.0 μW, below two linear states (104.6 μW)
- **Cold start:** 5–17% of reference; run-up advantage grows 1% → 29% from Q = 15 to 60

8 Limits and next steps

- **All results are computational:** single-DOF; no travel stops, misalignment, multi-axis vibration, temperature drift or higher modes
- PZ closed forms leading-order (up to 7.3% in the integral at Q = 10); tolerances studied for EM only; raw resistive-load power
- **Next:** test predicted state counts and loads on calibrated EM and PZ prototypes across a speed sweep

Key references

[1] Beeby, Tudor, White, Meas. Sci. Technol. 17 (2006) R175.
[2] Stephen, J. Sound Vib. 293 (2006) 409–425.
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[4] Langley, J. Sound Vib. 333 (2014) 927–936.
[5] Panter and Dite, Proc. IRE 39 (1951) 44; Gray and Neuhoff, IEEE Trans. Inf. Theory 44 (1998) 2325.
[6] Shu and Lien, Smart Mater. Struct. 15 (2006) 1499.
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[8] MaFaulDa (UFRJ); CWRU Bearing Data Center, normal baseline.
[9] Daqaq et al., Appl. Mech. Rev. 66 (2014) 040801.

Data: public recordings. Full bibliography (39 sources) in the paper.